Arithmetic Sequences and Series

ANSWER KEY

Arithmetic sequences

1 Consider the sequence $7, 11, 15, 19, \ldots$

a State the common difference. $d = 4$

b Write a formula for the nth term. $a_n = 4n + 3$

c Find a_{20}. $a_{20} = 83$

2 In an arithmetic sequence, $a_3 = 10$ and $a_8 = 25$.

a Find the common difference and the first term. $5d = 15$ gives $d = 3$; $a_1 = 10 - 2(3) = 4$.

b Write a formula for a_n. $a_n = 4 + 3(n-1) = 3n + 1$.

Arithmetic series

3 Find the sum of the first 30 terms of the series $5 + 9 + 13 + \cdots$

$a_{30} = 5 + 29(4) = 121$, so $S_{30} = \frac{30(5+121)}{2} = 1890$.

4 Consider the multiples of 6 strictly between 100 and 500.

a How many are there? First 102, last 498: $n = \frac{498-102}{6} + 1 = 67$.

b Find their sum. $S = \frac{67(102+498)}{2} = 67 \times 300 = 20{,}100$.

5 A theatre has 18 seats in the first row, and each subsequent row has 3 more seats than the row before. How many seats are there in total in 25 rows?

$a_{25} = 18 + 24(3) = 90$, so $S_{25} = \frac{25(18+90)}{2} = 1350$ seats.

Arithmetic means and sequence properties

6 Insert three arithmetic means between 5 and 25 (i.e., find b_1, b_2, b_3 so that $5, b_1, b_2, b_3, 25$ is arithmetic).

5 terms total, so $d = \frac{25-5}{4} = 5$. Means: $10, 15, 20$.

7 Given the arithmetic sequence with $a_1 = -8$ and $d = 3$, find the smallest n such that $a_n > 100$.

$a_n = -8 + 3(n-1) = 3n - 11 > 100$: $3n > 111$, $n > 37$: smallest integer $n = 38$.

8 Prove that the sum of the first n positive odd integers $(1 + 3 + 5 + \cdots)$ equals n^2, using the arithmetic series formula.

The nth odd number is $a_n = 2n - 1$, with $a_1 = 1$, $d = 2$. $S_n = \frac{n(a_1+a_n)}{2} = \frac{n(1+2n-1)}{2} = \frac{n(2n)}{2} = n^2$.

Applications

9 A person saves 100 riyals in week 1, and increases savings by 20 riyals each subsequent week. Find the total saved after 20 weeks.

$a_{20} = 100 + 19(20) = 480$. $S_{20} = \frac{20(100 + 480)}{2} = 5800$ riyals.

10 A stack of logs has 20 logs in the bottom row, 19 in the next, decreasing by 1 each row, ending with 1 log in the top row. Find the total number of logs.

This is the sum $1 + 2 + \cdots + 20 = \frac{20(1 + 20)}{2} = 210$ logs.